Q1 - 25 July - Shift 1

Following statements are given :

(1) The average kinetic energy of a gas molecule decreases when the temperature is reduced.

(2) The average kinetic energy of a gas molecule increases with increase in pressure at constant temperature.

(3) The average kinetic energy of a gas molecule decreases with increases in volume.

(4) Pressure of a gas increases with increase in temperature at constant pressure.

(5) The volume of gas decreases with increase in temperature.

Choose the correct answer from the options given below :

(A) (1) and (4) only (B) (1), (2) and (4) only

(C) (2) and (4) only (D) (1), (2) and (5) only

Space for your notes:

Q2 - 25 July - Shift 2

Sound travels in a mixture of two moles of helium and n moles of hydrogen. If rms speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound, then the value of n will be

(A) 1 (B) 2

(C) 3 (D) 4

Space for your notes:

Q3 - 26 July - Shift 2

A gas has n degrees of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be :

(A) $\frac{n}{n+2}$ (B) $\frac{n+2}{n}$

(C) $\frac{n}{2n+2}$ (D) $\frac{n}{n-2}$

Space for your notes:

Q4 - 27 July - Shift 1

Same gas is filled in two vessels of the same volume at the same temperature. If the ratio of the number of molecules is 1:4, then

A. The r.m.s. velocity of gas molecules in two vessels will be the same.

B. The ratio of pressure in these vessels will be 1 : 4

C. The ratio of pressure will be 1 : 1

D. The r.m.s. velocity of gas molecules in two vessels will be in the ratio of 1 : 4

(A) A and C only (B) B and D only

(C) A and B only (D) C and D only

Space for your notes:

Q5 - 27 July - Shift 2

η (first case) η (second case)

Which statements are correct about degrees of freedom?

A. A molecule with n degrees of freedom has n^2 different ways of storing energy.

B. Each degree of freedom is associated with $\frac{1}{2}RT$ average energy per mole.

C. A monoatomic gas molecule has 1 rotational degree of freedom where as diatomic molecule has 2 rotational degrees of freedom

D. CH_4 has a total to 6 degrees of freedom

Choose the correct answer from the option given below:

(A) B and C only (B) B and D only

(C) A and B only (D) C and D only

Space for your notes:

Q6 - 28 July - Shift 1

Given below are two statements :

Statement I :The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become 2v.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Space for your notes:

Q7 - 28 July - Shift 2

A vessel contains 14 g of nitrogen gas at a temperature of 27°C. The amount of heat to be transferred to the gap to double the r.m.s. speed of its molecules will be : (Take R = 8.32 J $mol^{-1}k^{-1}$)

(A) 2229 J (B) 5616 J

(C) 9360 J (D) 13,104 J

Space for your notes:

Q8 - 28 July - Shift 2

At a certain temperature, the degrees of freedom per molecule for gas is 8. The gas performs 150 J of work when it expands under constant pressure. The amount of heat absorbed by the gas will be …….. J.

Space for your notes:

Q9 - 29 July - Shift 1

The pressure P_1 and density d_1 of diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes suddenly to $P_2(>P_1)$ and d_2 respectively during an adiabatic process. The temperature of the gas increases and becomes __________ times of its initial temperature.

(given $\frac{d_2}{d_1} = 32$)

Space for your notes:

Q10 - 29 July - Shift 1

One mole of a monoatomic gas is mixed with three moles of a diatomic gas. The molecular specific heat of mixture at constant volume is $\frac{\alpha^2}{4}R$ J/mol K; then the value of α will be ________. (Assume that the given diatomic gas has no vibrational mode.)

Space for your notes:

Q11 - 29 July - Shift 2

The root mean square speed of smoke particles of mass 5×10^{-17} kg in their Brownian motion in air at NTP is approximately. [Given $k = 1.38 \times 10^{-23}$ JK^{-1}]

(A) 60 mm s^{-1} (B) 12 mm s^{-1}

(C) 15 mm s^{-1} (D) 36 mm s^{-1}

Space for your notes:

Answer Key

Q1 (A)	**Q2** (B)	**Q3** (A)	**Q4** (C)
Q5 (B)	**Q6** (D)	**Q7** (C)	**Q8** (750)
Q9 (4)	**Q10** (3)	**Q11** (C)	

Q1 (A)

$$KE_{avg} = \frac{3}{2}KT$$

$$P = \frac{1}{3}\rho V_{rms}^2$$

Note : Statement (4) is correct only if we consider it at constant volume and not constant pressure. Ideally, this question must be bonus but most appropriate answer is option (A)

Q2 (B)

$$v_s = \sqrt{\frac{\gamma RT}{M}}$$

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\frac{v_s}{v_{rms}} = \sqrt{\frac{\gamma}{3}} = \frac{1}{\sqrt{2}} \Rightarrow \frac{\gamma}{3} = \frac{1}{2} \Rightarrow \gamma = \frac{3}{2}$$

$$\gamma = 1 + \frac{2}{f_{mix.}}$$

$$f_{mix.} = \frac{2\times 3 + n\times 5}{n+2} = \frac{6 + n\times 5}{(n+2)}$$

$$\gamma = 1 + \frac{2(n+2)}{6 + n\times 5} = \frac{6 + 5n + 2n + 4}{6 + 5n}$$

$$\gamma = \frac{7n+10}{6+5n} = \frac{3}{2}$$

$$14n + 20 = 18 + 15n$$
$$n = 2$$

Q3 (A)

$$C_v = \frac{nR}{2} \qquad C_p = \frac{(n+2)R}{2}$$

$$\frac{C_v}{C_p} = \frac{n}{n+2}$$

Q4 (C)

KTG

A. $V_{Rms} = \sqrt{\frac{3RT}{M_w}} \Rightarrow V_{Rms}$ is same

B. $\frac{P_1}{P_2} = \frac{N_1}{N_2} \Rightarrow$ B is correct

Ans [A & B only are correct]

Q5 (B)

Methane molecule is tetrahedron

Degree of freedom due to rotation = 3

Degree of freedom due to translation = 3

Q6 (D)

$[P_{avg} = 0]$ (due to random motion)

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$T_{new} = 2T$$

$$M_{new} = \frac{M}{2}$$

$$\frac{v_{new}}{v} = \frac{\sqrt{\frac{2T}{M/2}}}{\sqrt{\frac{T}{M}}}$$

$$v_{new} = 2v$$

Q7 (C)

$$v_{rms} \propto \sqrt{T}$$

$$v_{rms} \propto \sqrt{300K}, \; v_{rms_f} = 2v_{rms_i}$$

$$v_{rms_f} \propto \sqrt{1200K}$$

$$T_f = 1200K, \; T_i = 300K, \; n = \frac{14}{28} = \frac{1}{2}$$

$$Q = nC_v\Delta T = \frac{1}{2} \times \frac{5R}{2} \times 900$$

$$Q = 9360 \text{ J}$$

Q8 (750)

$$W = nR\Delta T = 150 \text{ J}$$

$$\mathbf{Q} = \left(\frac{f}{2}+1\right) nR\Delta T = \left(\frac{8}{2}+1\right)150 = 750 \text{ J}$$

Q9 (4)

$$PV^{\gamma} = \text{const} \qquad d = \frac{m}{v}$$

$$p\left(\frac{m}{d}\right)^{\gamma} = \text{const}$$

$$\frac{p}{d^{\gamma}} = \text{const} \qquad \frac{d_2}{d_1} = 32$$

$$\frac{p_1}{p_2} = \left(\frac{d_1}{d_2}\right)^{\gamma} = \left(\frac{1}{32}\right)^{7/5} = \frac{1}{128}$$

$$\frac{T_1}{T_2} = \frac{P_1V_1}{P_2V_2} = \frac{1}{128}32 = \frac{1}{4}$$

Q10 (3)

$$C_V / mix = \frac{n_1 Cv_1 + n_2 Cv_2}{n_1 + n_2}$$

$$= \frac{1.\frac{3R}{2} + 3.\frac{5R}{2}}{1+3}$$

$$= \frac{9R}{4} = \frac{\alpha^2}{4} R$$

$$\alpha = 3$$

Q11 (C)

$$V_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 293}{5 \times 10^{-17}}}$$

$$\approx 15 mm / s$$